

TUNKU ABDUL RAHMAN UNIVERSITY OF MANAGEMENT AND TECHNOLOGY

FACULTY OF COMPUTING AND INFORMATION TECHNOLOGY

ACADEMIC YEAR 2025/2026

DECEMBER/JANUARY EXAMINATION

AMCS2123 SYSTEMS AND PROGRAMMING CONCEPTS

MONDAY, 5 JANUARY 2026

TIME: 2.00 PM – 4.00 PM (2 HOURS)

DIPLOMA IN COMPUTER SCIENCE

Instructions to Candidates:

Answer **ALL** questions. All questions carry equal marks.

AMCS2123 SYSTEMS AND PROGRAMMING CONCEPTS**Question 1**

- a) In what circumstance the **vector array** will be recommended for use? Show an example in C++ on how to declare a vector array called *numbers* of integer type and how an integer 50 can be added into the *numbers* array. (2 marks + 4 marks)

- b) With reference to the following table which is used to store the sales amount obtained for the six types of national cars throughout the 4 quarters of 2025, write C++ program statements to perform the following tasks: -

	1 st quarter 25	2 nd quarter 25	3 rd quarter 25	4 th quarter 25
MyVi				
Saga				
Alza				
Pesona				
Ativa				
X50				

- (i) Declare a 2-dimensional array for the table above. (2 marks)
- (ii) Change the sales of the car “Ativa” in 3rd quarter to RM295000. (2 marks)
- (iii) Calculate and display the total sales made in 4th quarter for all the above cars. (7 marks)
- (iv) Provide an example of a situation and reason where a nested *for* loop is used in the two-dimensional array. (3 marks)
- c) Explain the difference between array in structure and array of structure. Support your answer using C++ codes. (5 marks)
- [Total: 25 marks]

Question 2

- a) Write C++ statements to accomplish the following tasks:
- (i) Define a structure **Expenses** which consists of double type members for the different type of meal expenses spent for **breakfast**, **lunch** and **dinner**. Each of these meal expenses stores the expenses for 7 days of the week. Note that the first meal expenses stored for each type of meal refers to the expenses spent on Monday, the same goes to second meal expenses and so on. (3 marks)
- (ii) Declare a variable **expenses** of the Expenses structure type. (2 marks)
- (iii) Assume all the expenses in the variable **expenses** are filled up. Change the expenses for **lunch** spent on **Thursday** to 12.95. (2 marks)
- (iv) Define a structure **Siblings** with members **name** of string type and a nested structure **mealExpenses** of Expenses type. (2 marks)
- (v) Declare an array variable **siblings** of Siblings type for 5 siblings of the Chan family. (2 marks)

AMCS2123 SYSTEMS AND PROGRAMMING CONCEPTS**Question 2 a) (Continued)**

- (vi) Add 5.85 to the **dinner** expenses of **Saturday** for the **second sibling** in the Chan family. (4 marks)
- b) Perform the following tasks using C++ statements:
- (i) Prepare a template in C++ to display the temperature of the day of the week. The sample message to display is "Monday average temperature is 30.5 Celsius". (5 marks)
- (ii) Write a main function, which allows the user to enter the day of the week and the temperature, and then displays the message using the template prepared in Question 2 b) (i) above. (5 marks)
- [Total: 25 marks]

Question 3

- a) Through the use of recursive function, find the value of sum in the following expression by drawing the activation frames.
- $$\text{sum} = 1 + 2 + 3$$
- (5 marks)

- b) Show the output of the following C++ program statements:

```
int *pPtr, *qPtr;
int p, q, s;
pPtr = &p;
qPtr = &q;
*pPtr = 12;
q = 4;
cout << "p = " << p << "\nq = " << *qPtr << endl;
s = *pPtr * q;
cout << "s = " << s << endl;
(*pPtr) = *pPtr / 2;
q = *qPtr + 10;
cout << "p = " << p << "\nq = " << q << endl;
```

(5 marks)

- c) (i) Given a raw product code ABC1234567 which the program needs you to convert it into this meaningful code ABC-1234(567). Write a C++ program statement using string function to perform the conversion and save the result into a variable *code*. (4 marks)
- (ii) What is the use of *regex_search* function? (1 mark)
- d) Identify the storage class given the statements below:
- (i) The variables are not destroyed even if the function terminates and thus able to retain values between function calls. (1 mark)
- (ii) The variables are defined outside any functions and made available to any functions from the point they are declared. (1 mark)

AMCS2123 SYSTEMS AND PROGRAMMING CONCEPTS**Question 3 (Continued)**

- e) (i) Determine the bit pattern of 1011 1001 after left shift 2 bits. (1 mark)
- (ii) Determine the result of bit pattern for operation 1001 0110 | 0101 0010. (1 mark)
- f) (i) Give ONE (1) reason why tight coupling is not good. (1 mark)
- (ii) What level of coupling is the best one? (1 mark)
- (iii) Which level of cohesion is the best one and give the reason. (2 marks)
- g) Give **TWO (2)** benefits of modular programming. (2 marks)
- [Total: 25 marks]

Question 4

- a) Write a C++ boolean type function **processNumber** which receives an integer **num**, two integer type parameters **doubleVal** and **tripleVal** using **pass by reference** and **pass by address** respectively. The function should calculate the double ($\text{num} * 2$) and triple ($\text{num} * 3$) of **num**. It should also check whether **num** is **positive**. The function returns true if **num** is positive, otherwise false. (8 marks)

- b) Use *typedef* to define new type names **Day**, **Month** and **Year** of integer type. Then write a C++ program that does the necessary declaration using the new type names above and prompts the user to enter their birth date consisting of **day**, **month**, and **year**. After collecting the inputs from the user, display the full birth date in the format below:

Date of birth: day/month/year

Note: The day, month and year in the format above are user input values. (9 marks)

- c) Given the following structure declaration for **Student**:-

```
struct Student {
    string name, id;
    int pspMark;
};
Student stud = {"Henry Ford", "25WMR12345", 86};
string result;
```

- (i) Write a C++ function **checkResult** which receives a mark of integer type, and it will return the **result** of string type using pass by address. The function will check if the mark is greater and equal to 50, it will assign "PASS" to the result, otherwise "FAIL". (6 marks)
- (ii) Construct a function call statement to perform the **checkResult** function using the data of the variable **stud** provided above. (2 marks)

[Total: 25 marks]